`is-cref` — Icelandic declension for cleveref

v0.1

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Abstract

The `is-cref` package adds Icelandic grammatical declension to cross-references produced by `cleveref`. It allows references to figures, tables, equations, and sections to appear in the correct grammatical case — nominative, accusative, dative, or genitive — in both singular and plural, with automatic case selection from prepositions.

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1 Introduction

The `cleveref` package automates cross-reference formatting in LaTeX but has no Icelandic language support and can only produce the nominative case. Icelandic is a heavily inflected language with four grammatical cases, and reference nouns such as *mynd* (figure), *tafla* (table), *jafna* (equation), and *kafli* (section) must be declined according to their grammatical role in the sentence. `is-cref` solves this by storing declension tables for each reference type and allowing the user to specify the case via a simple optional argument.

Icelandic has four cases:

- **Nominative** (*nefnifall*) — subject position
- **Accusative** (*þolfall*) — direct object, some prepositions
- **Dative** (*þágufall*) — indirect object, most locative prepositions
- **Genitive** (*eignarfall*) — possession, some prepositions

2 Requirements and loading

`is-cref` automatically loads the following packages:

- `hyperref`
- `cleveref`
- `xparse`

Load the packages in the following order in your preamble:

```
\usepackage[hidelinks]{hyperref}   % or colorlinks=true
\usepackage{cleveref}
\usepackage{is-cref}
```

3 Main commands

3.1 \icref

```
\icref[<case>]{<label>}
```

Produces a declined reference noun followed by the number, with the number hyperlinked to its target. The optional `<case>` argument accepts the following inputs:

Input	Case	Example
(none)	nominative	`\icref{fig:a}`
`nom` or `nf` or `0`	nominative	`\icref[nom]{fig:a}`
`acc` or `þf` or `1`	accusative	`\icref[acc]{fig:a}`
`dat` or `þgf` or `2`	dative	`\icref[dat]{fig:a}`
`gen` or `ef` or `3`	genitive	`\icref[gen]{fig:a}`
preposition	automatic	`\icref[í]{fig:a}`

When a preposition is given, the command outputs the preposition followed by the declined noun:

```
\icref[í]{fig:a}       % → í mynd 1
\icref[vegna]{fig:a}   % → vegna myndar 1
\icref[um]{eq:a}       % → um jöfnu (1)
```

If an unrecognised argument is passed, the package issues a warning and falls back to the nominative.

3.2 `\Icref`

```
\Icref[<case>]{<label>}
```

Identical to `\icref` but produces a capitalised form for use at the start of a sentence:

```
\Icref{fig:a}        % → Mynd 1
\Icref[dat]{fig:a}   % → Mynd 1
```

3.3 Plural references

Both `\icref` and `\Icref` handle plurals automatically when more than one label is given:

```
\icref{fig:a,fig:b}         % → myndir 1 og 2
\icref[dat]{fig:a,fig:b}    % → myndum 1 og 2
```

Numerically consecutive labels are compressed automatically:

```
\icref{eq:a,eq:b,eq:c}      % → jöfnur (1) til (3)
```

4 Declaring reference types

4.1 `\DeclareIcrefType`

```
\DeclareIcrefType{<type>}
  {nom sg}{acc sg}{dat sg}{gen sg}
  {nom pl}{acc pl}{dat pl}{gen pl}
```

Declares the eight declined forms for a reference type. `<type>` is the internal cleveref counter name — `figure`, `table`, `equation`, `section`, etc.

```
\DeclareIcrefType{figure}
  {mynd}{mynd}{mynd}{myndar}
  {myndir}{myndir}{myndum}{mynda}
```

4.2 \DeclareIcrefTypeCapital

Identical syntax to `\DeclareIcrefType` but declares the capitalised forms used by `\Icref`:

```
\DeclareIcrefTypeCapital{figure}
  {Mynd}{Mynd}{Mynd}{Myndar}
  {Myndir}{Myndir}{Myndum}{Mynda}
```

Both commands should be called together for every custom type. If `\DeclareIcrefTypeCapital` is not called, `\Icref` will fall back to `\Cref` with a package warning.

5 Preposition table

5.1 \DeclareIscrefPreposition

Users can add new prepositions or override existing ones:

```
\DeclareIscrefPreposition{<preposition>}{<case>}
```

Where `<case>` is `acc`, `dat`, or `gen`. For example, to override `"á"` to accusative:

```
\DeclareIscrefPreposition{á}{acc}
```

5.2 Built-in prepositions

Case	Prepositions
Accusative	um, gegnum, kringum, umfram, umhverfis
Dative	frá, að, af, andspænis, ásamt, gagnvart, gegn, gegnt, handa, hjá, meðfram, nálægt, undan, úr, [á, í, fyrir, undir, yfir, við, með, eftir]*
Genitive	til, auk, austan, án, handan, innan, meðal, megin, milli, millum, neðan, norðan, ofan, sakir, sunnan, sökum, utan, vegna, vestan

* These prepositions can govern either accusative or dative depending on context (motion vs. location). Dative is the default as locative readings are most common in technical documents. This can be overridden with `\DeclareIscrefPreposition`.

6 Built-in reference types

The following types are built in with full declension tables, singular:

Type	Nom.	Acc.	Dat.	Gen.
`figure`	mynd	mynd	mynd	myndar
`table`	tafla	töflu	töflu	töflu
`equation`	jafna	jöfnu	jöfnu	jöfnu
`section`	kafli	kafla	kafla	kafla

and plural:

Type	Nom.	Acc.	Dat.	Gen.
`figure`	myndir	myndir	myndum	mynda
`table`	töflur	töflur	töflum	taflna
`equation`	jöfnur	jöfnur	jöfnum	jafna
`section`	kaflar	kafla	köflum	kafla

7 Complete example

```
\documentclass{article}
\usepackage[icelandic]{babel}
\usepackage[hidelinks]{hyperref}
\usepackage{cleveref}
\usepackage{is-cref}

\begin{document}

\begin{figure}
  \caption{Yfirlit}
  \label{fig:yfirlit}
\end{figure}

Niðurstöður eru sýndar í \icref{fig:yfirlit}.
Fjallað er um \icref[acc]{fig:yfirlit} í næsta kafla.
Sjá má frekari upplýsingar \icref[í]{fig:yfirlit}.
\Icref{fig:yfirlit} sýnir samantekt á gögnum.

\end{document}
```

8 Acknowledgements

This package builds on `cleveref` by Toby Cubitt. The preposition system was inspired by `crefthe` and `xcref`.